Market Sentiment & Momentum: Evaluating Predictive Signals across the Magnificent Seven

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Introduction

Forecasting short-term stock returns is one of the harder problems in quantitative finance, and for good reason; daily prices reflect the aggregate judgment of millions of participants responding to overlapping streams of information in real time. This project does not attempt to solve that problem. Instead, it takes an exploratory look at a narrower question: does the sentiment of publicly available news coverage carry any measurable relationship with next-day price behavior for the Magnificent Seven technology stocks?

The MAG7, which consists of Apple, Microsoft, Google, Amazon, Nvidia, Meta, and Tesla, sits at the intersection of high news volume and high market visibility, making it a natural starting point for a study like this. We sourced 89,019 articles from the GDELT news database across a two-year window (February 2024 – February 2026), scored each headline using FinBERT, a transformer-based model trained specifically on financial text, and aligned the resulting sentiment signals to next-trading-day OHLCV price data. After aggregation, the working dataset contains more than 1,300 ticker-day observations.

Our analysis is both hypothesis-driven and exploratory. We entered the project with four specific claims about how sentiment might relate to returns, volatility, and momentum, and we followed the data honestly from there. The results are descriptive and correlational; no claims of causality are made, and nothing here constitutes financial advice or a trading strategy. What we hope to demonstrate is that a carefully constructed pipeline, from raw news ingestion through transformer-based sentiment scoring to structured model evaluation, can surface meaningful patterns in a domain where clean answers are rare.

Research Questions

This project is organized around four driving research questions:

1. Do sentiment scores vary meaningfully across Magnificent Seven companies?
2. Are negative sentiment signals associated with increased short-term risk or volatility?
3. Does sentiment have stronger relationships with same-day or next-trading-day market movement?
4. How sensitive are our results to data source coverage and article volume?

Data & Pipeline

Sources

The analysis draws on two primary data sources: news articles collected through the GDELT Document 2.0 API and daily OHLCV price data for each of the seven tickers.

GDELT (Global Database of Events, Language, and Tone) is a free, open-platform news index that monitors print, broadcast, and web sources in over 100 languages. We used the Doc 2.0 API to collect articles mentioning each MAG7 company by name over the two-year period, totaling 89,019 articles. It is worth noting that GDELT over-represents free, web-accessible outlets. Paywalled sources such as the Wall Street Journal, Bloomberg, and the Financial Times are underrepresented, which means the sentiment signal is best understood as a measure of free-web news coverage rather than institutional financial press. This is an inherent ceiling on the sentiment signal.

Daily OHLCV price data (open, high, low, close, adjusted close, and volume) were collected for all seven tickers across the same window. Price data was validated against the NYSE market calendar to account for trading holidays and weekends, and cleaned for logical errors and outliers prior to joining.

Pipeline

Raw data passed through a multi-stage pipeline before reaching any analysis: Ingestion → Validation Processing Accumulation Features Modeling Evaluation.

During validation, duplicate URLs and near-identical headlines were removed, non-English articles were filtered out, and malformed records were dropped. Cleaned articles were built up across multiple ingestion runs into a stable, deduplicated dataset. All transformations are deterministic and write to a separate processed directory; raw data is treated as immutable. The environment is fully specified in environment.yml, and all outputs can be regenerated from committed code.

Sentiment Scoring - FinBERT

Each article headline was scored using FinBERT, a BERT-based transformer model fine-tuned on financial news. FinBERT produces a three-class probability distribution (positive, negative, neutral) for each input. Neutrality rates are high across headlines, consistent with prior work on financial NLP: most financial news is factual and does not use strong directional language. Only articles that cleared a sentiment threshold of $|score| > 0.05$ contributed a non-zero score to downstream aggregation.

An earlier version of the pipeline used a custom word-bank lexicon (v1) for sentiment scoring. FinBERT (v2) replaced it due to substantially better coverage and contextual sensitivity. The two scoring methods are not directly comparable in magnitude and should not be treated interchangeably.

News-Price Join

A core design decision in the pipeline is the temporal alignment of news to prices. An article published on day t is joined to the OHLCV record for the next trading day (t+1), with weekend and holiday articles rolling forward to the following Monday or post-holiday session. News published during or after market hours is assumed not to be reflected in prices until the following open. Each article carries its sentiment score into the joined record and is aggregated at the ticker-day level. After joining and removing ticker-days with insufficient coverage, the working dataset consists of more than 1,300 ticker-day observations spanning the full two-year window.

Pipeline Overview

Raw News (GDELT)

→Validation & Cleaning

→Sentiment Scoring (v2: FinBERT)

→Aggregation (Ticker-Day Level)

→Feature Engineering

→Modeling (Momentum vs Sentiment)

→Evaluation (TimeSeriesSplit & Walk-Forward)

Feature Engineering

With the joined dataset in hand, the next step was to construct the features used for modeling. The central concern throughout was leakage, the risk that a feature inadvertently encodes information about the future, producing results in development that would be impossible to replicate in practice. Every price-based feature was built with a strict minimum shift of 1 day, meaning each feature at time t draws only on data available at the close of day t-1 or earlier.

mom_5d captures five-day price momentum for each ticker, measuring the direction and magnitude of recent price movement in the days leading up to the observation.

volume_z is a rolling z-score of daily volume, capturing unusual trading activity relative to each ticker's own recent history. Abnormal volume has been associated in prior literature with short-term return continuation and reversal effects.

neg_extreme is a binary indicator that flags ticker-days in the bottom decile of sentiment scores. Rather than treating sentiment as a continuous predictor, where the signal-to-noise ratio is poor, this feature captures the tail of the distribution where sentiment may have a more concentrated effect.

momentum × neg_extreme (interaction term) is the central feature of interest in the modeling phase. It tests whether extreme negative sentiment alters the relationship between momentum and next-day returns, rather than acting as an independent predictor in its own right.

A note on one feature that was ultimately excluded: a one-day momentum feature (mom_1d) was initially included but removed after being identified as a source of leakage. Because mom_1d was computed using the same-day close, it encoded same-day return information that would not be available when a prediction was needed. All results reported in this paper use the zero-leakage feature set.

Methodology

Exploratory Analysis

Hypothesis Testing

The first phase of the project was structured around four explicit hypotheses about the relationship between sentiment and returns. Rather than exploring freely, we committed to specific, testable claims upfront and evaluated each against the data.

Hypotheses:

- **H1** - Sentiment score has a measurable correlation with next-day returns.
- **H2** - Sentiment is a lagging indicator of price movement; news reacts to returns rather than driving them.
- **H3** - Extreme negative sentiment is associated with elevated next-day volatility or downside risk.
- **H4** -Sentiment modifies the relationship between momentum and next-day returns
- **H5** - Robustness checks, walk-forward validation, stress-testing of H4 (rejected)

Interpretive Extensions:

- **I1/H6** - Sentiment acts as a **regime-based** modifier, influencing how momentum behaves under different market conditions.

- Each hypothesis was evaluated using Spearman's correlation, OLS regression, and logistic regression, as appropriate.

- Statistical significance was assessed using both standard p-values and cluster-robust (HC1) standard errors grouped by date, to account for the fact that observations on the same trading day are not independent.

- Robustness was tested across multiple sentiment threshold definitions to avoid anchoring conclusions to any single specification; ultimately, all four hypotheses were ultimately rejected under the feature set and out-of-sample validation conditions.

Modeling Approach

The second phase extended the hypothesis testing into a full classification pipeline. The prediction target is beat_market, a binary label indicating whether a given ticker outperformed the MAG7 average return on a given day. A leave-one-out market return is used to avoid target leakage, since a naive average of all seven tickers would encode the ticker's own return.

Baseline: A momentum-only model (forward_return ~ momentum) establishing the performance floor.

Advanced: forward_return ~ momentum + sentiment + (momentum × sentiment). Adds sentiment features and the interaction term to test whether sentiment provides incremental predictive value beyond momentum alone.

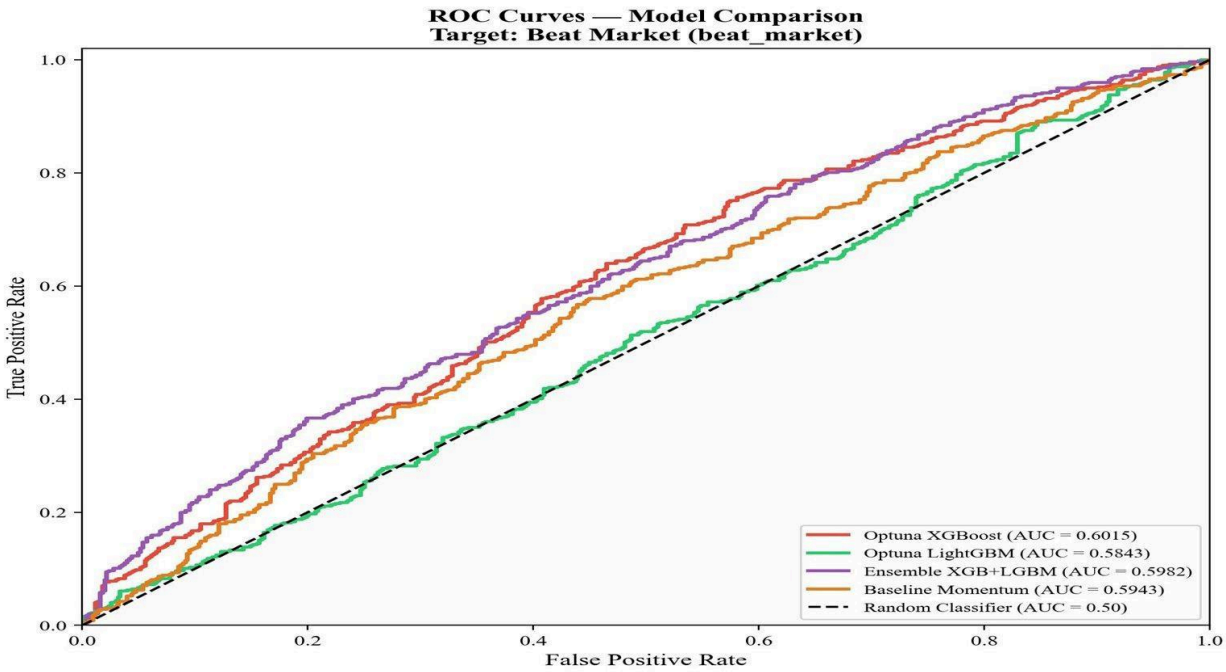
Classification: P(beat_market) estimated using tree-based models (RandomForest, LightGBM, XGBoost, GradientBoosting) and logistic regression. All models were evaluated using 5-fold TimeSeriesSplit cross-validation to respect the temporal ordering of the data, with a separate held-out test set and walk-forward validation used to assess true out-of-sample performance.

Across all modeling approaches, including tree-based models and tuned ensembles, predictive performance consistently plateaued in the ~58-61% accuracy range, suggesting a natural ceiling given the available signal and dataset size *after filtering for sufficient MAG7 inclusion*.

Table 1. Model Performance Comparison - Accuracy and AUC

Model Name	Features	ACC	AUC	Notes
LogReg	Top Features	0.560	0.566	Baseline model with simple linear relationships
LogReg	Full Features	0.543	0.554	Adding all features reduced performance, likely introduced noise
Random Forest	Full Features	0.581	0.590	Captures non-linear relationships, moderate improvement
XGBoost (Conservative)	Full Features	0.587	0.582	Strong performance, improves generalization over Random Forest
XGBoost (Conservative)	Top Features	0.594	0.570	Feature selection improves stability but slightly lowers AUC
LGBM (Conservative)	Full Features	0.590	0.579	Comparable to XGBoost, efficient boosting method
XGBoost (Tuned)	Full Features	0.550	0.607	Hyperparameter tuning improves AUC but lowers ACC
LGBM (Tuned)	Full Features	0.592	0.606	Best balance of ACC and AUC after tuning
Ensemble (XGBoost + LGBM, Tuned)	Full Features	**0.598**	**0.607**	Best overall performance (highest AUC with strong ACC); ensemble improves robustness
Ensemble (XGBoost + LGBM, Optuna Tuned)	Expanded Features	0.608	0.598	Slight accuracy gain, but reduces AUC; new features likely added noise or overfitting
Walk-Forward XGBoost	Full Features	0.554	0.586	Stable performance in time-based validation; realistic performance

Figure 1. ROC Curves - Model Comparison (Target: beat_market)



Key Findings

Sentiment Alone

Across all four hypotheses, raw sentiment showed no consistent predictive power for next-day returns. Correlations were weak, unstable across different threshold definitions, and failed to reach significance under robust testing conditions. Treating sentiment as a standalone signal does not hold up in this dataset.

This may reflect structural limitations in the dataset. Financial news is predominantly neutral in tone, particularly given that we were applying our scoring to headline text exclusively.

GDELT's bias toward free web sources excludes high-signal, paywalled institutional reporting, and, as a result, sentiment signals are both sparse and noisy, limiting their predictive power.

Momentum

Short-term momentum (primarily 5-day momentum) emerged as the strongest and most reliable signal throughout the analysis, with a Spearman correlation of $\rho = +0.167$ ($p < 0.0001$) against the beat_market target. This is consistent with the broader financial literature on short-term return continuation in high-liquidity equities. Momentum is the dominant signal in this study.

Extreme Sentiment

Isolating the bottom decile of sentiment scores, the neg_extreme flag, did not produce a reliable independent effect on returns. While extreme negative sentiment showed some association with downside behavior in certain specifications, the effects were inconsistent across time periods and sensitive to the definition of the extreme threshold. No standalone predictive claim can be made.

Interaction (Key Insight)

The most notable in-sample finding is that extreme negative sentiment appears to modify the relationship between momentum and next-day returns, rather than acting as a predictor on its own. Under normal sentiment conditions, momentum behaves as expected. Under extreme negative sentiment, the relationship shifts. However, this interaction weakened meaningfully under cluster-robust (HC1) standard errors grouped by date, and showed near-zero out-of-sample R^2 under walk-forward validation. The finding is directionally interesting but not sufficiently stable to be considered a validated signal.

Exploratory regime-based analysis suggests that this interaction may vary across sentiment-defined conditions, with momentum exhibiting different behavior under extreme sentiment regimes; however, these effects were not consistently stable during out-of-sample validation and are therefore treated as *interpretive* rather than predictive.

Validation & Limitations

All hypotheses were evaluated both in-sample and out-of-sample. In-sample testing used OLS regression, logistic models, and marginal effects analysis. Out-of-sample performance was assessed using walk-forward validation, where models were trained on past data and evaluated on future periods to simulate realistic prediction conditions.

The key out-of-sample result is straightforward: sentiment does not provide reliable predictive improvement over momentum alone. Walk-forward R^2 was near zero, and the interaction term that appeared meaningful in-sample did not hold up under temporal validation.

Several limitations constrain the conclusions. Financial return data is inherently noisy at the daily frequency, making it difficult to isolate weak signals such as sentiment. Results were also sensitive to how sentiment thresholds were defined — conclusions shifted depending on whether a 10% or 20% extreme cutoff was used, or which version of the sentiment score was applied.

Finally, the interaction signal showed temporal instability, declining in later periods of the two-year window, suggesting that any effect may be regime-dependent rather than structural.

Capstone 1 Conclusion

Momentum is the primary predictive signal in this study. It is consistent, statistically significant, and robust across specifications. Sentiment, by contrast, is not a standalone predictor; it showed no reliable correlation with next-day returns across any of the four hypotheses tested.

The most defensible conclusion from this study is that momentum dominates short-term predictive performance in high-liquidity equities, while aggregated news sentiment does **not** provide a reliable incremental signal at a daily frequency.

While extreme sentiment may influence market behavior under certain conditions, these effects are not sufficiently stable to forecast. The primary contribution of this work is not the discovery of tradable signals, but the construction of a reproducible, end-to-end pipeline that rigorously evaluates these signals under realistic constraints.

Further work, such as more data, longer windows, or alternative sentiment sources, would be needed before any stronger claims could be made.

Team Contributions

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